

Algebra II with Trigonometry

Chapter 7.2

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Problems

1. Evaluate exactly: $\cos \left(\tan^{-1}(2) + \tan^{-1}(3) \right)$.
2. Find the exact value of $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$.
3. Compute the exact value of $\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ}$.
4. Find the exact value of $\sec^2 \left(\frac{1}{2} \sin^{-1} \left(\frac{4}{5} \right) \right) + \csc^2 \left(\frac{1}{2} \cos^{-1} \left(\frac{3}{5} \right) \right)$.
5. Evaluate exactly: $\sin^2 10^\circ + \sin^2 50^\circ + \sin^2 70^\circ$.
6. Compute the exact integer value of the product: $\prod_{k=1}^{59} (1 + \sqrt{3} \tan k^\circ)$.